

SURF-DUNE facility interface

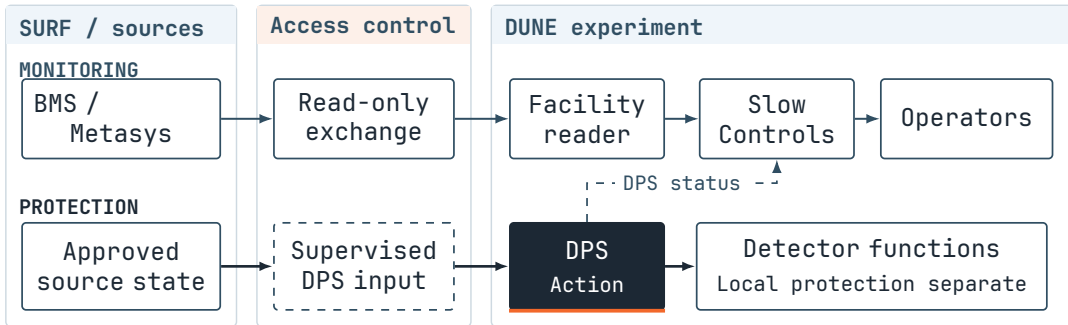
Building management system integration

8 September 2026

Fermilab · DUNE Far Detector · Underground facility, South Dakota

How facility signals reach DUNE

Proposed paths for monitoring and detector protection.



Functional view

Trip link to be defined.

Lost monitoring

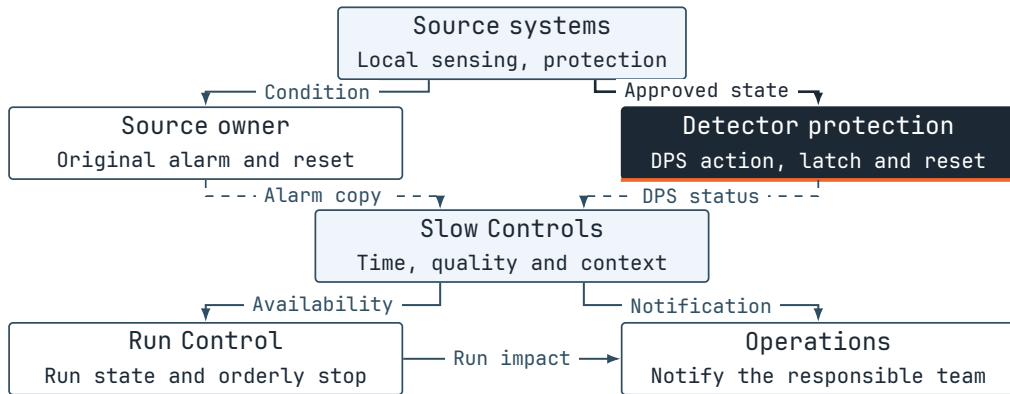
Data gaps are flagged.

Recovery

Owner confirms restart.

Alarm flow and ownership

Information moves through the experiment; authority stays with the owner.



DUNE acknowledges the display copy, not the source alarm or DPS reset.

Facility conditions

The source definition and agreed cause and effect determine the response.

CONDITION	POSSIBLE DUNE RESPONSE
Power / reserve	An agreed request for an orderly DAQ stop.
Network / BMS	Stale or unknown data; the loss is visible.
Fire / low oxygen	Source response; approved detector action.
LAr leak	Response follows a defined source condition.
Cooling / water	Local protection; agreed DPS action.
Weather / access	SURF restrictions inform operations.

Loss of monitoring is a design case for the independent protection path.

Proposed data exchange

Protocol support and source ownership remain to be confirmed.

INTERFACE	CANDIDATE	INFORMATION FLOW
BMS readout	BACnet/IP	Selected source points to reader.
DUNE view	OPC UA	Reader to Slow Controls.
Health	SNMPv3	Power and network telemetry.

Read requests and replies do not grant control authority.
BACnet/SC and SNMPv3 depend on support. The DPS interface is separate.

From concept to agreed interface

Three areas for discussion with the source and experiment teams.

1 Point meaning and ownership

Each point has a defined source, state, quality and owner.

2 Data path and timing

The interface describes access, update rate and data age.

3 Response and recovery

Checks cover alarms, loss, approved action and reset.
